

Project Design Phase-II Data Flow Diagram & User Stories

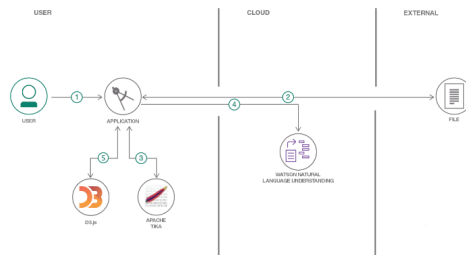
Date	05 July 2026
Team ID	
Project Name	Visualization tool for electric vehicle charge and range analysis.
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: (Simplified)

Flow

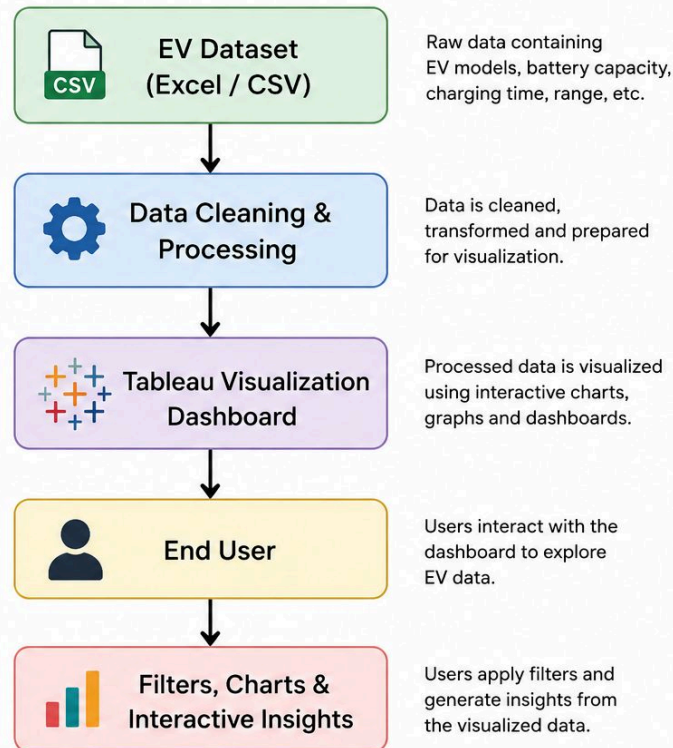


1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

Example: DFD Level 0

Data Flow Diagram (DFD) – Level 0

Visualization Tool for Electric Vehicle Charge and Range Analysis



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
User	View Dashboard	USN-1	As a user, I want to open the EV dashboard so that I can view electric vehicle charging and range information.	Dashboard loads successfully.	High	Sprint-1
User	Filter Data	USN-2	As a user, I want to filter the dashboard by vehicle model, manufacturer, or year to analyze specific data.	Filters work correctly and update the charts.	High	Sprint-1
User	Analyze Range	USN-3	As a user, I want to compare the driving range of different electric vehicles.	Range comparison charts display accurate values.	High	Sprint-1
User	Analyze Charging	USN-4	As a user, I want to view charging time and battery capacity information.	Charging-related visualizations are displayed correctly.	Medium	Sprint-1
User	Insights	USN-5	As a user, I want to identify trends and patterns in EV performance through visualizations.	Dashboard provides clear and interactive insights	High	Sprint-2
Administrator	Update Dataset	USN-6	As an administrator, I want to update the EV dataset so that the dashboard reflects the latest information.	Updated dataset refreshes the dashboard successfully.	Medium	Sprint-2